

QWIM MARKET REGIME DETECTION AND REGIME-AWARE PORTFOLIO ALLOCATION



Medium–Long Term Wealth Management Application

Phase 1

FE800

Stevens Institute of Technology

Group 1: Mina Fahim | Nabil Salame | Willie Gadson | Niti Bhavesh Shah | Ojaus Mane | Darshan Nanjegowda

Spring 2026

Why Regimes Matter for Medium–Long Term Wealth Allocation

- Financial markets are **not stationary**.
- Over time, we observe persistent structural shifts in:
 - Expected returns (*Nystrup et al., 2017*)
 - Volatility and tail risk (*Nystrup et al., 2017; Aydinhan et al., 2024*)
 - Cross-asset correlations (*Nystrup et al., 2017; Aydinhan et al., 2024*)
 - Credit conditions (*Aydinhan et al., 2024*)
 - Inflation (*Pomorski, 2024*)
 - Monetary and macro cycles (*Pomorski, 2024; Hirsä et al., 2024*)
- **Our thesis:** Portfolio allocation should adapt to persistent regime shifts, while controlling turnover and transaction costs.
 - Objective:
 - Improve drawdown resilience
 - Enhance smooth compounding
 - Preserve capital across cycles



Objective

- Build a comprehensive multi-asset feature library
- Capture market, macroeconomic, and cross-asset regime information
- Use the engineered dataset in Phase 2 for:
 - Hidden Markov Model
 - Semi-Hidden Markov Model
 - Jump / Continuous Jump Model
 - RHSM
 - Machine Learning model



Regimes We Aim to Identify ($K \in \{3,5\}$)

Regime 1

Risk-On / Expansion

- **Market Characteristics:**
 - Positive medium-term returns
 - Moderate volatility
 - Stable cross-asset correlation structure (no systemic spike)
 - Tightening credit spreads
- **Macro Characteristics:**
 - Growth stable or improving
 - Inflation contained
 - Neutral or easing policy
- **Portfolio Implication:**
 - Higher equity factor exposure
 - Increased cyclical (Value, Size)
 - Reduced cash

Regime 2

Risk-Off / Stress

- **Market Characteristics:**
 - Negative returns
 - Elevated volatility
 - Correlation spikes
 - Widening HY spreads
- **Macro Characteristics:**
 - Growth deterioration
 - Policy tightening or shock
 - Inflation instability possible
- **Portfolio Implication:**
 - Reduced equity exposure
 - Defensive tilt (Quality, Low Vol)
 - Increased IG and cash

Regime 3

Transitional / High-Uncertainty

(if $K = 3$)

- **Market Characteristics:**
 - Elevated volatility-of-volatility
 - Weak or unstable trend
 - Rising correlation dispersion
- **Macro Characteristics:**
 - Turning points in growth or inflation
 - Policy inflection
- **Portfolio Implication:**
 - Reduced conviction
 - Ambiguity-aware shrinkage toward neutral
 - Balanced exposure



Data Collection

Content

- **Period:** Dec 1999 – Dec 2025
- **Frequency:** Daily (market), Monthly (macro)
- **Sources**
 - **Bloomberg:** Market, rates, credit, macro data
 - **Kenneth French Library:** Momentum factor
- **Asset Coverage**
 - **Equities:** SPX, Value, Growth, Small Cap
 - **Fixed Income:** Treasury curve (2Y–30Y), IG & HY credit
 - **Commodities:** Brent Oil, Gold
 - **FX & Risk:** DXY, VIX
 - **Macro:** Inflation, Growth, Liquidity, Recession

Bloomberg



Data Collection – Asset Coverage

- **Equity Markets**

- **SPX Index** | broad market
- **RLV Index** | Value
- **RU20INTR Index** | Growth
- **RTY Index** | Small caps

- **Fixed Income & Credit**

- **LUACTRUU Index** | Investment Grade Bonds
- **LF98TRUU Index** | High Yield Bonds

- **Interest Rates (Treasury Curve)**

- **GT2 Govt** | 2-Year Treasury Yield
- **GT10 Govt** | 10-Year Treasury Yield
- **GT30 Govt** | 30-Year Treasury Yield

- **Credit Spreads**

- **LUAS Index** | Investment Grade Spread
- **LF98OAS Index** | High Yield Spread

- **Commodities**

- **Brent Crude Oil** | Global growth / inflation proxy
- **GOLDLNPM Index** | safe-haven asset

- **FX & Risk Indicators**

- **DXY Index** | U.S. Dollar strength
- **VIX Index** | Implied equity volatility

- **Macro & Economic Indicators**

- **CPI Index, CPURNSA Index** | Inflation
- **CONSENT Index, CONCCONF Index** | Consumer sentiment
- **ISM Manufacturing, ISM Services** | Economic activity
- **M1, M2** | Liquidity / money supply
- **NBER Recession Indicator** | Business cycle

- **Factor Data**

- **Fama-French Momentum Factor**



Data Collection – Portfolio Universe

- **Equity ETFs**

- **SPY** | Broad U.S. Equity
- **IVE** | Value
- **IVW** | Growth
- **IWM** | Small Cap

- **Alternative ETF**

- **GLD** | Gold

- **Fixed Income ETFs**

- **SHY** | Short-Term Treasuries (2Y)
- **IEF** | Intermediate Treasuries (10Y)
- **TLT** | Long-Term Treasuries (30Y)
- **LQD** | Investment Grade Credit
- **HYG** | High Yield Credit

ETF	Benchmark (Market Signal)
SPY	SPX Index
IVE	RLV Index (Value)
IVW	RU20INTR Index (Growth)
IWM	RTY Index (Small Cap)
SHY	GT2 Govt
IEF	GT10 Govt
TLT	GT30 Govt
LQD	LUACTRUU Index
HYG	LF98TRUU Index
GLD	GOLDLNPM Index



Data Preparation

All features are transformed into a consistent, time-aligned dataset suitable for regime detection.

- Steps
 - Collect raw market and macro series
 - Align data by date
 - Convert all numeric fields
 - Engineer regime-relevant features
 - Export feature buckets into processed datasets
- Timing Convention for Macroeconomic and survey variables
 - Macro variables are not observed in real time
 - To prevent look-ahead bias, we apply a **one-period lag**:

$$X_t^{macro} = X_{t-1}^{macro}$$



Feature Engineering Overview

We construct 45+ features across key economic dimensions:

- Equity Market Dynamics
- Volatility & Risk Sentiment
- Interest Rate Environment
- Cross-Asset Risk Signals
- Global Risk Conditions
- Economic Activity
- Inflation Environment
- Liquidity & Monetary Conditions
- Safe Haven Signals
- Commodity & Inflation Shocks
- Bond Market Features
- Credit Market Stress

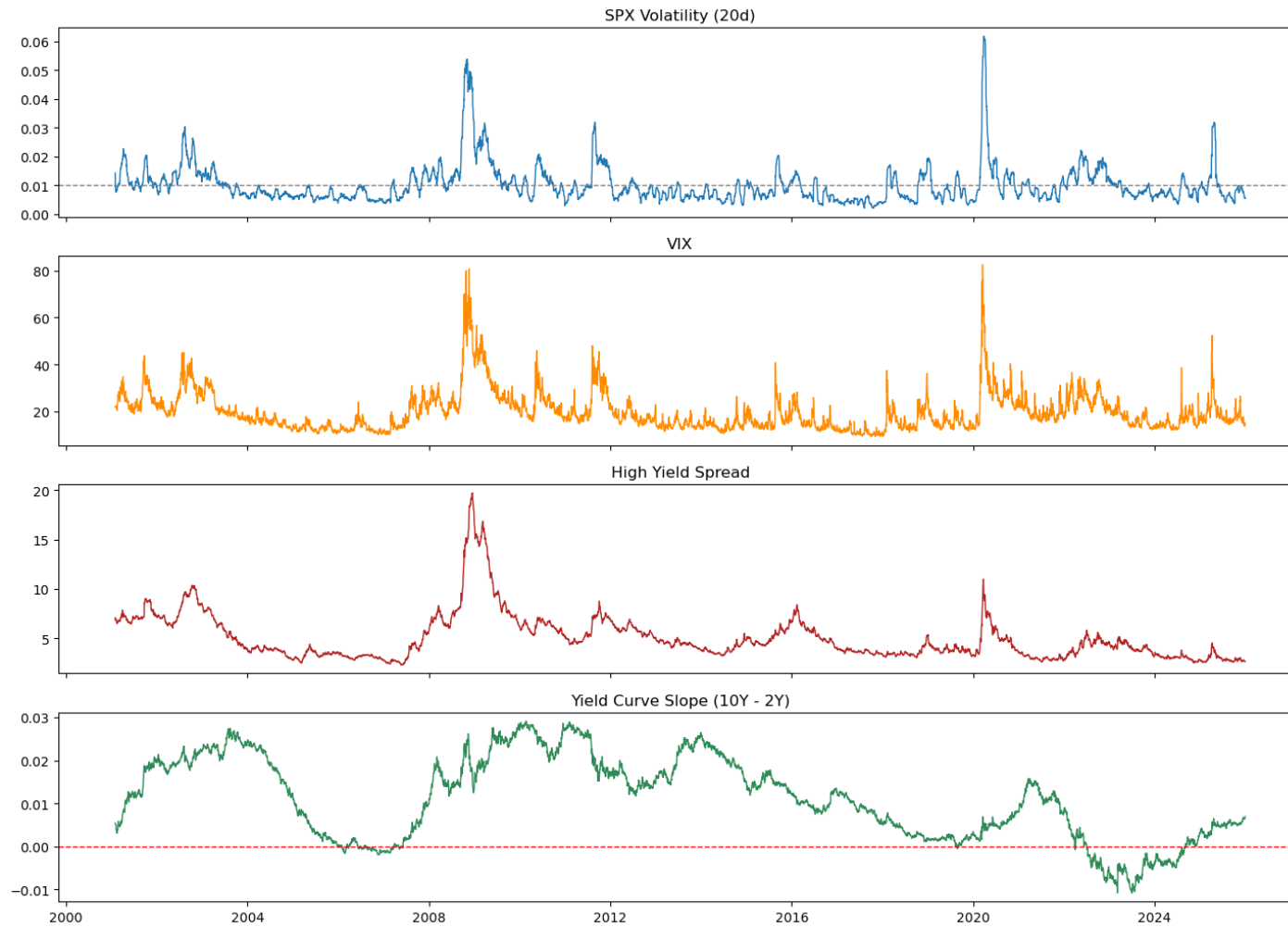


	Bucket	Feature_Name	Final_Name	Lagged	Frequency	Feature_Type	Use	Description
0	Equity Market Dynamics	SPX_return	SPX_return	No	Daily	Return	Core	$\log(\text{SPX}_t / \text{SPX}_{\{t-1\}})$
1	Equity Market Dynamics	SPX_volatility_20d	SPX_volatility_20d	No	Daily	Volatility	Core	rolling std (20d)
2	Equity Market Dynamics	SPX_drawdown	SPX_drawdown	No	Daily	Drawdown	Core	rolling max drawdown
3	Equity Market Dynamics	SPX_MA_signal	SPX_MA_signal	No	Daily	Trend	Optional	MA50 - MA200
4	Equity Market Dynamics	Momentum Factor	Mom	No	Daily	Factor	Optional	Fama-French momentum
5	Volatility / Risk	VIX_level	VIX	No	Daily	Level	Core	Raw VIX
6	Volatility / Risk	VIX_change	VIXChange	No	Daily	Change	Core	log change
7	Volatility / Risk	VIX_term_structure	VIX3M_minus_VIX	No	Daily	Spread	Optional	VIX3M - VIX
8	Rates	Slope_10_2	Slope_10_2	No	Daily	Spread	Core	10Y - 2Y
9	Rates	Slope_30_10	Slope_30_10	No	Daily	Spread	Optional	30Y - 10Y
10	Rates	YieldChange	YieldChange	No	Daily	Change	Core	GT10 - lag(GT10)
11	Rates	YieldCurvature	YieldCurvature	No	Daily	Curvature	Optional	2*10Y - 2Y - 30Y
12	Cross Asset	Eq-Bond Spread	Return_Eq_Bond	No	Daily	Spread	Core	SPX - Bond
13	Cross Asset	Eq-Bond Corr	Equity_bond_corr	No	Daily	Correlation	Core	rolling corr
14	FX	DXY_return	DXY_return	No	Daily	Return	Optional	log return
15	FX	DXY_momentum	DXY_mom_63	No	Daily	Momentum	Optional	63d
16	FX	DXY_vol	DXY_vol_21	No	Daily	Volatility	Optional	21d std
17	FX	DXY_SPX_corr	DXY_SPX_corr	No	Daily	Correlation	Optional	corr
18	Macro Growth	Industrial_ISM	Industrial_ISM_lag1	Yes	Monthly→Daily	Level	Core	Lagged ISM
19	Macro Growth	ISM_change	ISM_change_lag1	Yes	Monthly→Daily	Change	Optional	diff + lag
20	Inflation	CPI_change	CPI_change_lag1	Yes	Monthly→Daily	Change	Optional	diff + lag
21	Inflation	CPI_yoy	CPI_yoy_lag1	Yes	Monthly→Daily	YoY	Core	YoY inflation
22	Inflation	CPI_change_3m	CPI_change_3m_lag_1	Yes	Monthly→Daily	Smoothed	Optional	3m avg

	Bucket	Feature_Name	Final_Name	Lagged	Frequency	Feature_Type	Use	Description
23	Inflation	Inflation_Regime	Inflation_Regime_lag1	Yes	Monthly→Daily	Regime	Core	CPI > 3%
24	Inflation	UMich	UMich_Sentiment_lag1	Yes	Monthly→Daily	Level	Core	Lagged sentiment
25	Inflation	CB	CB_Confidence_lag1	Yes	Monthly→Daily	Level	Core	Lagged confidence
26	Liquidity	M2_growth	M2_growth_lag1	Yes	Monthly→Daily	Change	Optional	pct_change
27	Liquidity	M2_growth_3m	M2_growth_3m_lag1	Yes	Monthly→Daily	Smoothed	Optional	3m
28	Liquidity	M2_growth_yoy	M2_growth_yoy_lag1	Yes	Monthly→Daily	YoY	Core	YoY
29	Safe Haven	Gold_return	GLD_Return	No	Daily	Return	Optional	log return
30	Safe Haven	Gold/SPX	GLD_SPX_Ratio	No	Daily	Ratio	Core	Gold / SPX
31	Safe Haven	Gold/SPX Vol	GLD_SPX_Vol_21	No	Daily	Volatility	Optional	21d std
32	Commodity	Oil_return	Oil_Return	No	Daily	Return	Optional	log return
33	Commodity	Oil_vol	Oil_Vol_21	No	Daily	Volatility	Optional	21d
34	Commodity	Oil_mom	Oil_Mom_63	No	Daily	Momentum	Optional	63d
35	Credit	IG Spread	IG_Spread	No	Daily	Level	Core	IG spread
36	Credit	HY Spread	HY_Spread	No	Daily	Level	Core	HY spread
37	Credit	HY Change	HY_Change	No	Daily	Change	Core	diff
38	Credit	IG Change	IG_Change	No	Daily	Change	Optional	diff
39	Label	NBER_Recession	NBER_Recession	No	Monthly	Label	Evaluation	Do NOT use in ML

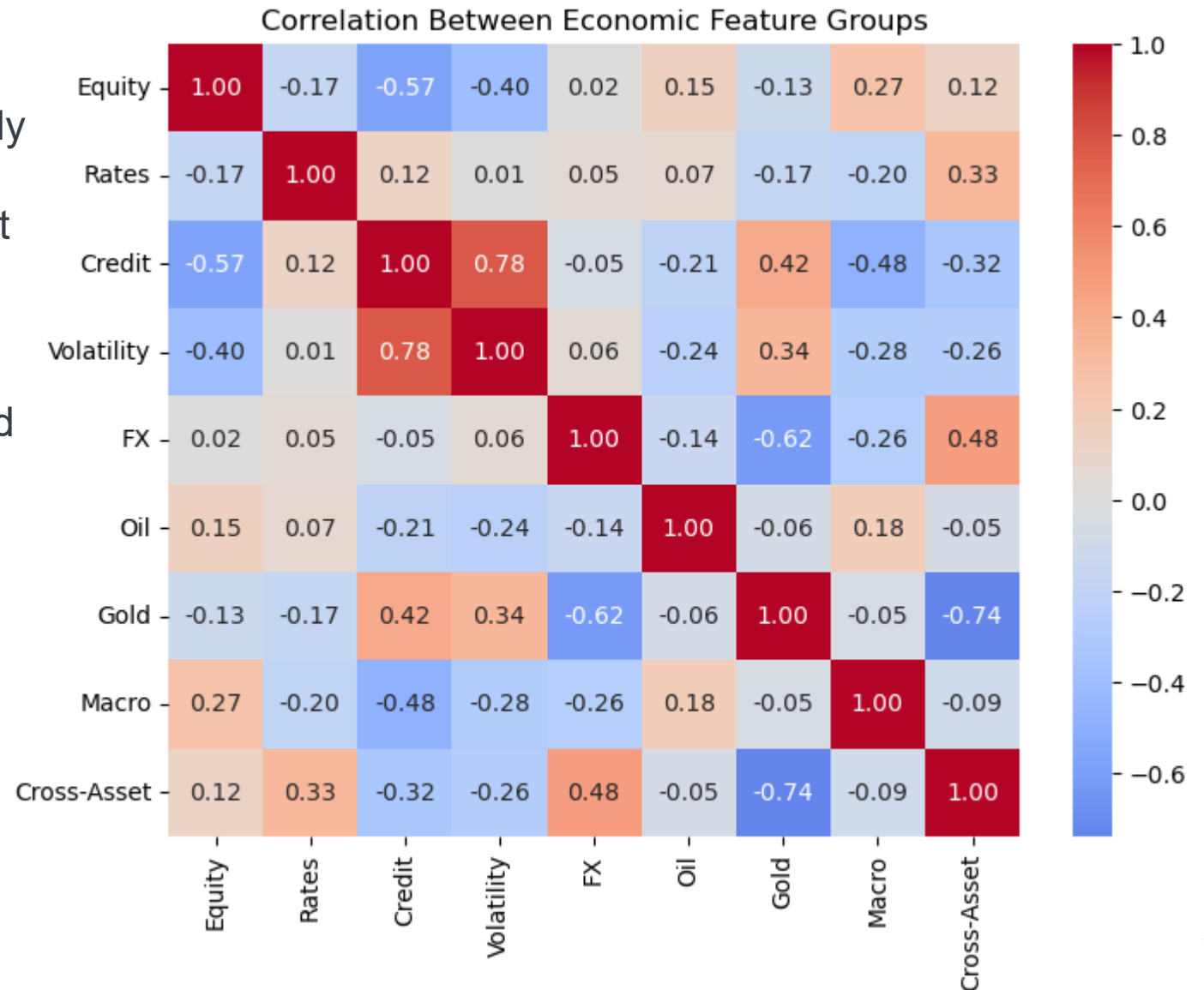


Representative Feature Dynamics

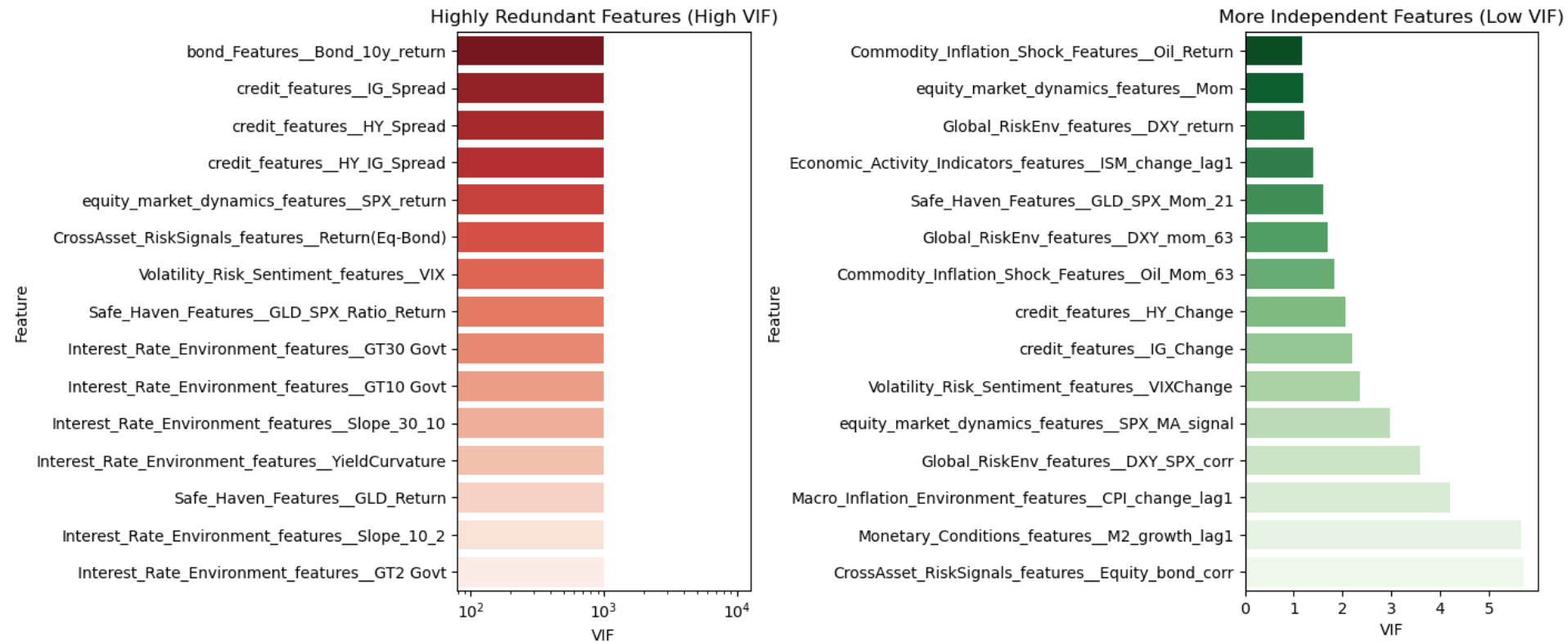


Correlation Between Feature Buckets

- Credit and volatility are highly positively correlated
- Equity negatively correlated with credit and volatility
- Strong correlation within risk-related features
- Gold negatively correlated with FX and cross-asset signals
- Macro variables show weaker and slower relationships

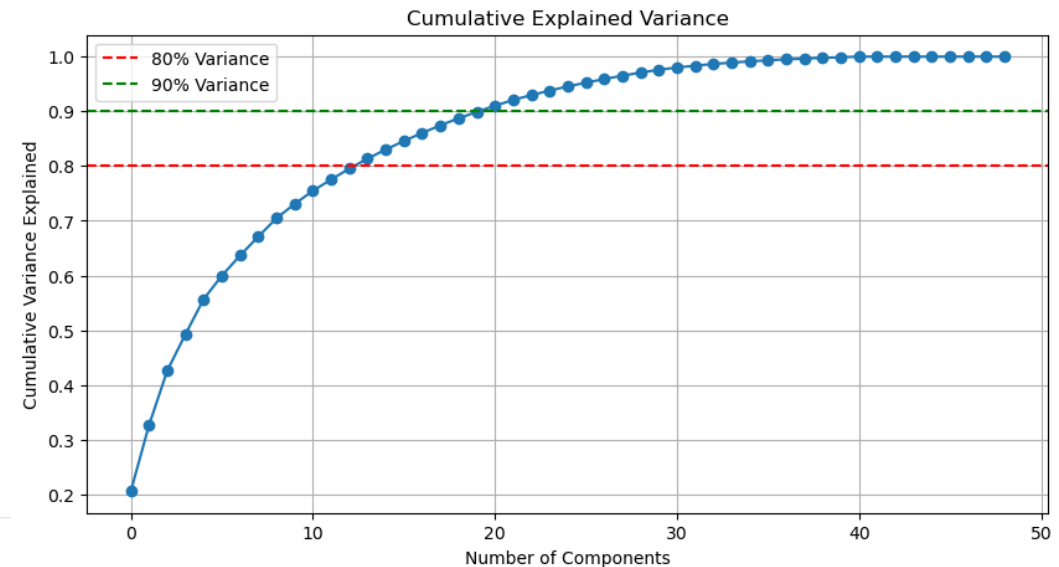
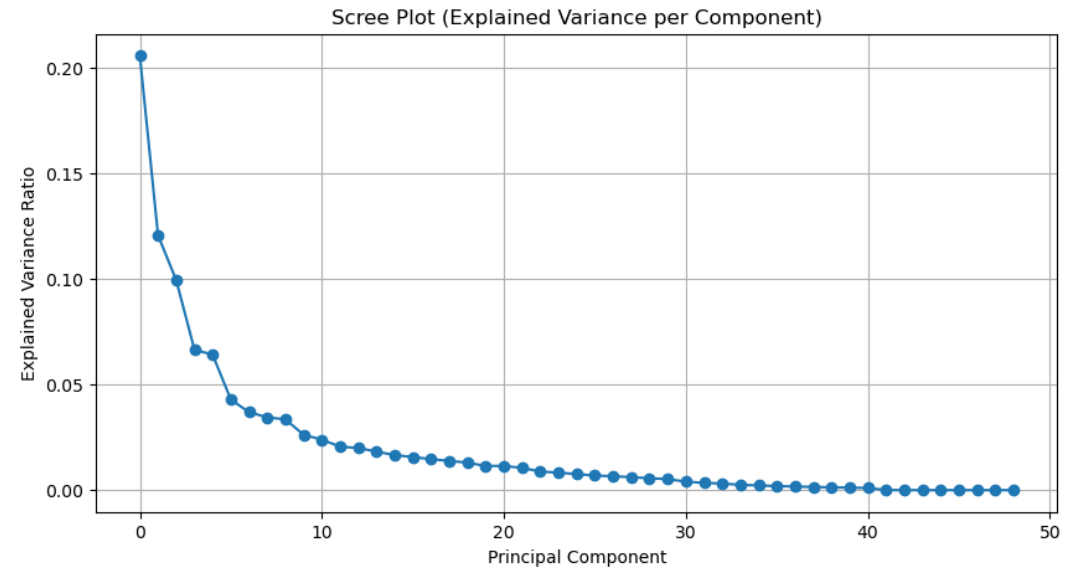


Variance Inflation Factor Analysis – Multicollinearity

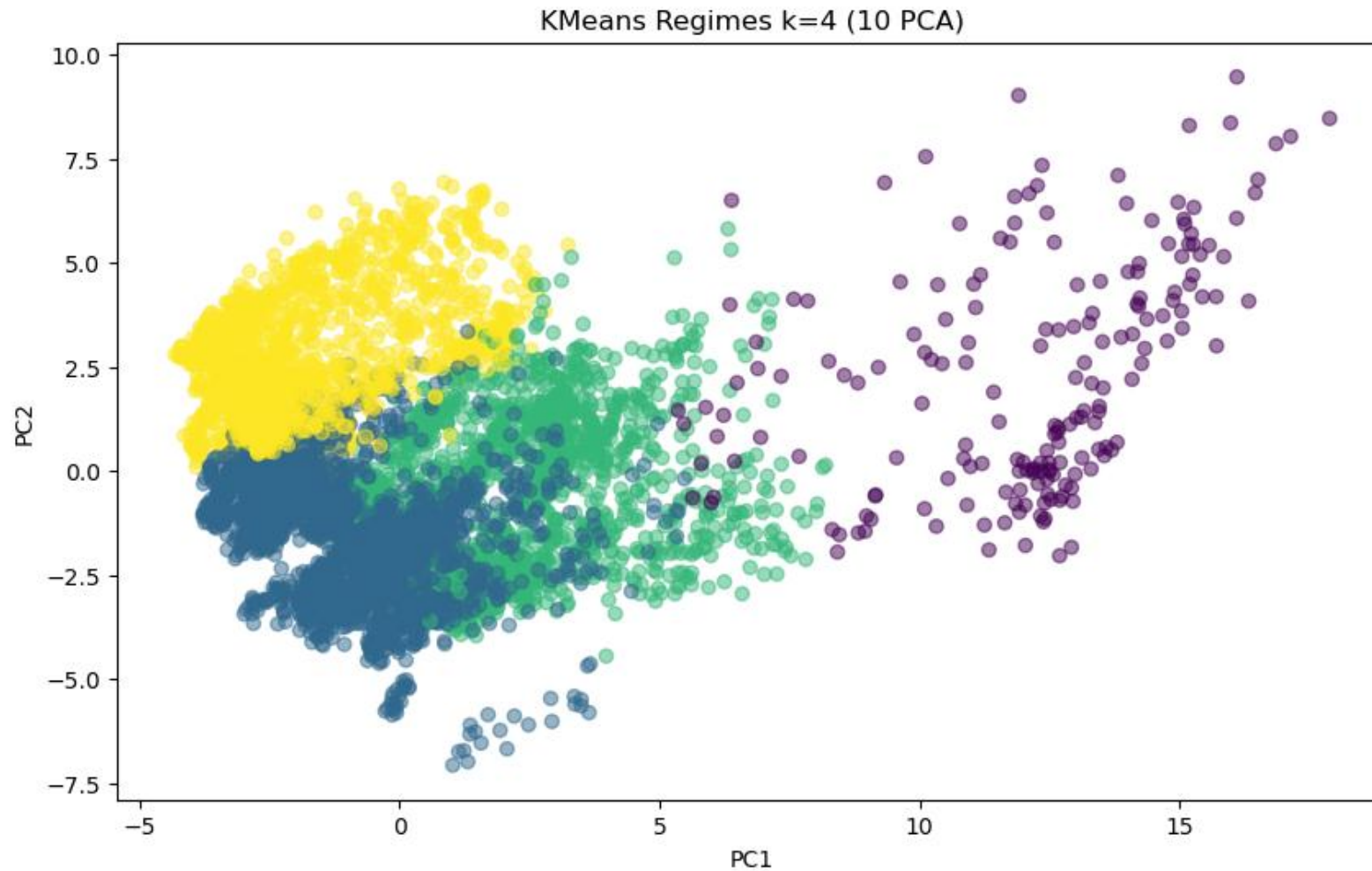


Principal Component Analysis (PCA)

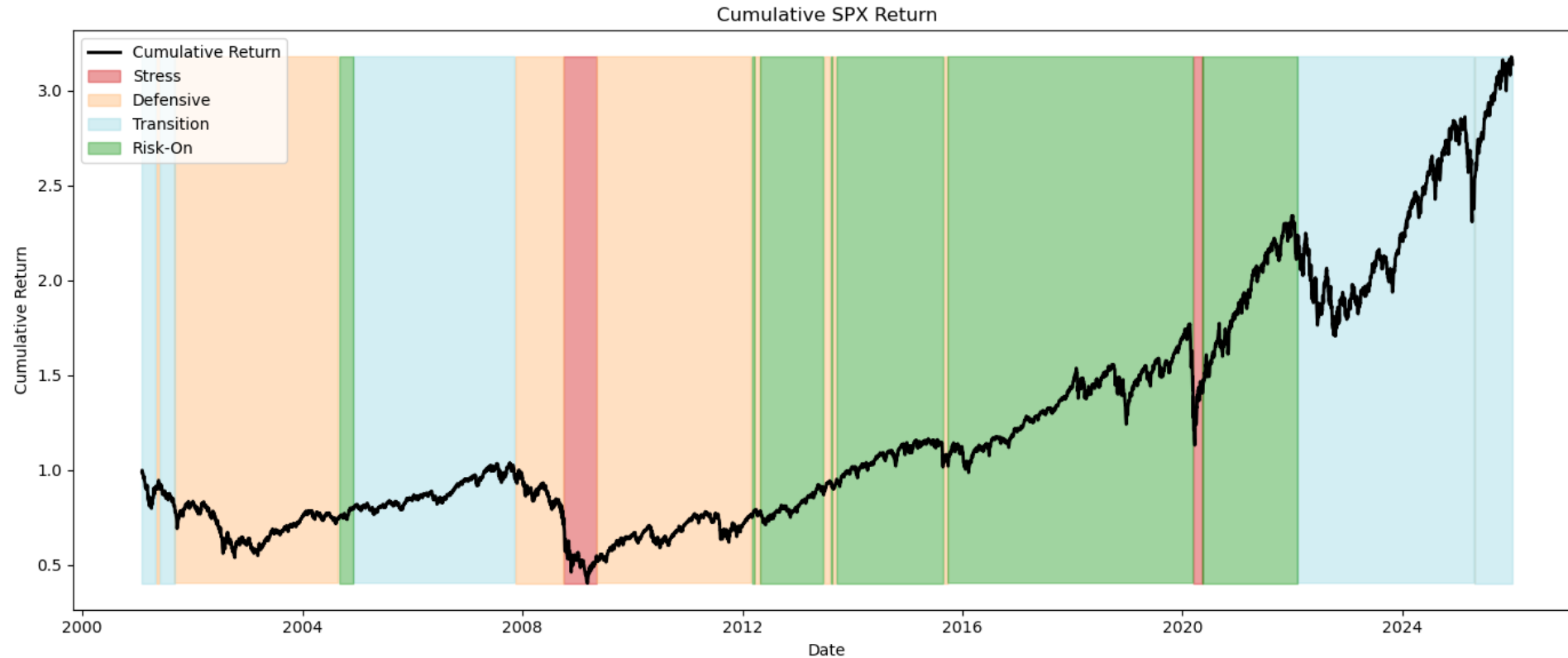
- Features are standardized before PCA
- PCA reduces dimensionality while preserving structure
- First 8-10 components explain most of the variance
- Additional components contribute less information
- Supports a lower-dimensional representation of market states



Regime Detection (K-Means Clustering)



Regime Validation & Interpretation



Next Steps:

- **Regime Detection Models**
 - Implement **HMM**, **Semi-HMM**, **Jump Models**, **RHSM**, and **ML** approaches
 - Use existing implementations where available
 - Develop custom models where needed
- **Model Robustness & Validation**
 - Test different numbers of regimes
 - Evaluate stability and persistence
 - Perform out-of-sample validation
- **Portfolio Construction**
 - Build regime-aware allocation framework
 - Mean-variance optimization conditional on regimes
 - Incorporate transaction costs and turnover
- **Dashboard & Visualization**
 - Develop interactive dashboard for regime monitoring
 - Visualize regime probabilities and transitions
 - Track portfolio performance across regimes





THANK YOU

Stevens Institute of Technology
1 Castle Point Terrace, Hoboken, NJ 07030